Weather-Based Activity Agent

Innovative Assignment 1 – Micro Level Project
Course: Foundations of Artificial Intelligence
Project Type: Rational Agent

1. Problem Formulation

The Weather-Based Activity Agent is a rule-based rational agent that recommends a suitable activity based on current weather conditions.

The user provides three environmental inputs: temperature in degrees Celsius, weather condition, and wind level. The agent observes these inputs, applies predefined decision rules, and recommends either an indoor or outdoor activity.

The percepts are temperature, weather condition, and wind level. The action is an activity recommendation. The goal is to select a safe and suitable activity.

The agent follows the cycle: Perceive -> Decide -> Act.

2. Approach

The project uses a rule-based decision-making approach. The recommend_activity() function receives temperature, weather condition, and wind level and evaluates them using conditional rules.

Rainy, stormy, or thunderstorm weather results in an indoor recommendation. Temperatures of 35°C or above result in an indoor recommendation. High wind also results in an indoor recommendation.

Sunny or clear weather with suitable temperature and low or moderate wind results in an outdoor recommendation. Cloudy or partly cloudy conditions result in an outdoor recommendation. Unknown conditions use a safe indoor fallback.

The program also validates user input. Missing weather information, invalid temperature values, and invalid wind levels are handled with clear messages instead of allowing the program to crash.

3. Algorithm

1. Start the program. 2. Read temperature from the user. 3. Validate the temperature. 4. Read the weather condition. 5. Check for missing weather data. 6. Read the wind level. 7. Validate the wind level. 8. Apply the weather decision rules. 9. Display the recommended activity. 10. End the program.

4. Complexity Analysis

The agent checks a fixed number of decision rules sequentially. If k represents the number of rules, the decision process has O(k) time complexity. Since k is fixed in this implementation, the practical time complexity is O(1).

Only a small number of input variables and temporary values are stored, so the space complexity is O(1).

5. Results and Testing

The agent was tested using normal weather, extreme conditions, unknown weather, and invalid inputs.

Manual tests included sunny weather producing an outdoor recommendation, rainy weather producing an indoor recommendation, high temperature producing an indoor recommendation, high wind producing an indoor recommendation, and unknown weather producing a safe fallback.

Invalid temperature, missing weather, and invalid wind input were also tested. The program handled these cases without crashing.

Five automated test cases were created using Python's unittest framework. The automated test suite completed successfully with: Ran 5 tests — OK.

6. Evidence / Screenshots

The following screenshots provide evidence of the implemented agent and its testing.

Screenshot 1

```
@bhuvanesh4127 →/workspaces/weather-activity-agent (main) $ python src/weather_agent.py
     WEATHER-BASED ACTIVITY AGENT
==================================
Enter temperature (°C): 28
Enter weather condition: sunny
Enter wind level (low/moderate/high): low

Agent Decision:
Outdoor activity recommended: Go for a walk, cycling, or play outdoor games.
@bhuvanesh4127 →/workspaces/weather-activity-agent (main) $ python src/weather_agent.py
```

Screenshot 2

```
@bhuvanesh4127 →/workspaces/weather-activity-agent (main) $ python src/weather_agent.py
Outdoor activity recommended: Go for a walk, cycling, or play outdoor games.
@bhuvanesh4127 →/workspaces/weather-activity-agent (main) $ python src/weather_agent.py
==================================
     WEATHER-BASED ACTIVITY AGENT
==================================
Enter temperature (°C): 25
Enter weather condition: rainy
Enter wind level (low/moderate/high): moderate
```

7. Limitations

The current version requires weather information to be entered manually by the user. It does not currently obtain live weather information from an API. Recommendations are based on predefined rules and the number of activities is limited.

8. Future Improvements

The project can be improved by connecting the agent to a live weather API, adding humidity and precipitation probability, providing more personalised recommendations, adding more weather conditions, and developing a graphical user interface.

9. Reflection

The main challenge was designing decision rules that produce sensible recommendations for different weather conditions. Another important part was handling invalid and missing user input so that the program does not crash.

The project demonstrates how a rational agent observes its environment, evaluates conditions, and selects an action according to a defined objective.

AI assistance and online/reference material were used during development. The project code and design should be understood by the team members and explained during the viva.

10. Conclusion

The Weather-Based Activity Agent demonstrates a simple rational agent using rule-based decision making. The agent receives weather-related information from the user, evaluates the conditions, and recommends an appropriate indoor or outdoor activity.

The project demonstrates the AI agent cycle: Perceive -> Decide -> Act. Manual and automated testing show that the implemented rules work for normal, extreme, unknown, and invalid input conditions.